

Farside Seismic Suite: Update on the status of the first seismic station on the farside of the Moon. M. P. Panning¹, Sharon Kedar¹, Asad Aboobaker¹, Glenn Aveni², Kevin Biernacki³, Neil Bowles⁴, Simon Calcutt⁴, Gabrielle Chabaud², Melanie Drilleau⁵, Taoufik Gabsi², Raphael Garcia⁵, Charlotte Giroud⁶, Alain Givaudan³, Anna Horleston⁷, Maurice Karakak³, Taichi Kawamura², Thelma Lestage⁸, Phillipe Lognonné^{2,9}, Josipa Majstorovic⁹, Ed Miller¹, David Mimoun⁵, Tanguy Nebut², Ceri Nunn¹, Sreejaya Kizhaekke Pakkathillam^{2,9}, Constanza Pardo², W. Tom Pike¹⁰, Gabriel Pont¹¹, Sebastien de Raucourt², Olivier Robert², Daniel Sheward¹², Sylvain Tillier², Renee Weber¹³, and Arnaud Wilhelm⁵ ¹Jet Propulsion Laboratory, California Institute of Technology, 4800 Oak Grove Dr., Pasadena, CA 91109, USA, ²Institut de Physique du Globe, Paris, France, ³Astroparticule et Cosmologie (APC), France, ⁴University of Oxford, UK, ⁵ISAE Supaero, France, ⁶ESTACA, France, ⁷University of Bristol, UK, ⁸Hensoldt Space Consulting, France, ⁹Université Paris Cité, France, ¹⁰Imperial College, London, UK, ¹¹CNES, France, ¹²Observatoire de la Côte d'Azur, France, ¹³NASA Marshall Space Flight Center

Introduction: The Farside Seismic Suite (FSS), selected in 2021 for flight as part of the NASA PRISM (Payloads and Research Investigations on the Surface of the Moon) program and planned for flight in 2026 on a lander provided by a group led by the commercial vendor Draper, would deliver two seismometers (both flight-proven through the InSight mission to Mars [1]) to Schrödinger Basin, as part of a package of payloads called CP-12 by NASA. The vertical Very BroadBand (VBB) seismometer, provided by CNES with design work from IPG, Paris, is the most sensitive flight-ready seismometer ever built [2], while the Short Period (SP) sensor, delivered by Kinemetrics, Inc. with design work at Imperial College, London and Oxford University, is the most sensitive and mature compact triaxial sensor available for space application [2]. FSS delivers a self-sufficient payload, with independent power, communications and thermal control allowing survival and operation over the long lunar night independent of the lander which it will outlive.

CP-12 also includes a magnetotelluric sounding system [3] and heat flow probe [4], meaning that CP-12 represents deployment of most key geophysical instrumentation proposed for the Lunar Geophysical Network [5]. While FSS is only originally proposed to last 4.5 months, the basic thermal design suggests a powerful path forward to implement a lunar seismic network using relatively small payloads with seismic instrumentation designed for multi-year survival delivered by affordable, short-lived landers.

FSS science is described in more detail in previous LPSC abstracts [e.g. 6]. In short, FSS was designed to meet 3 key science objectives (Fig. 1):

1. Investigate deep lunar structure and the difference between near and farside activity.
2. Understand how the lunar crust is affected by the development of an impact melt basin.
3. Evaluate the current micrometeorite impact rate and local tectonic activity at Schrödinger Basin.

Current Status: The FSS flight model is currently in final integration and testing (I&T). Environmental testing is expected to be completed in early 2024, after

which the flight unit will be kept in storage until integration on the lander, which is planned for 2025 to prepare for a launch to the Moon in 2026.

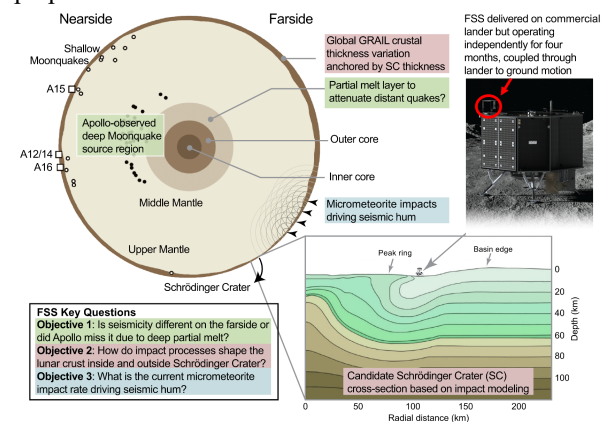


Figure 1. FSS will return data from Schrödinger Basin over multiple lunar diurnal cycles after outliving the delivery lander. FSS provides: (1) The first seismic data probing the deep lunar interior and determining the differences between near and farside activity (green); (2) Measurement of crustal thickness and layering that illuminates the process of crater formation (pink); (3) Current micrometeorite impact rate through recording of background seismic vibrations (blue). Figure adapted from [7] and [8]. Lander image courtesy of Draper Labs.

The FSS design consists of a cube within a cube (Figure 2, top). The inner cube houses all of the seismometers, the battery, the avionics and instrument electronics. The outer cube allows for mounting to the lander deck as well as supporting the solar array, radiator, and S band antenna for communications with a Draper-provided relay satellite. Between the cubes, several layers of spacerless multi-layer insulation (MLI) provides thermal control with a thermal switch and mini loop heat pipe linking the inner assembly to the radiator permitting both efficient heat transport during the hot lunar day and effective insulation during the cold lunar night. As of early January, 2024, the inner cube has been

fully assembled (Figure 2, middle), allowing for initial tests of system and instrument functionality as well as electromagnetic conductance and radiation testing (Figure 2, bottom).



Figure 2. (Top) CAD rendering of FSS design illustrating the cube within a cube design.. (Middle) Image of a partial flight assembly of the inner cube, clearly showing the flight instruments and battery pack. The completed inner cube assembly also includes avionics, instrument electronics, and a loop heat pipe and thermal switch. (Bottom) Completed inner cube assembly mounted on tilt fixture allowing for testing of seismometer with simulated lunar gravity along the sensing axis.

Upcoming I&T work: In the coming few months, the inner cube will be mounted within the outer cube using titanium tube supports, and the MLI, solar array, radiator, and radio antenna will be mounted. At that point, assembly will be complete, and vibration, as well as thermal and vacuum testing will be performed. Upon successful completion of testing and post-environmental functional verification, FSS will be moved into storage at JPL until Draper is ready for payload integration on the lander (anticipated in 2025).

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